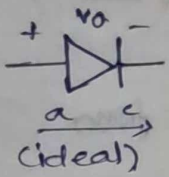


BASIC ELECTRONICS

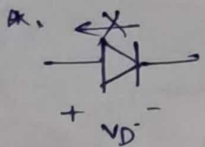
CIRCUIT ANALYSIS



→ two terminal device; non-linear device.

current flows from anode to cathode. (not reverse)

* $i_D > 0 \Rightarrow V_D = 0; R_D = 0 \Omega$

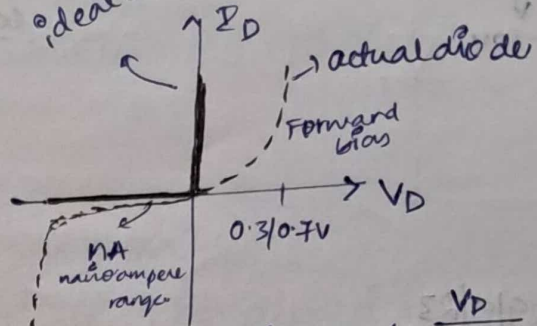


$i_D = 0$

$V_D < 0$

$R_D = \infty$

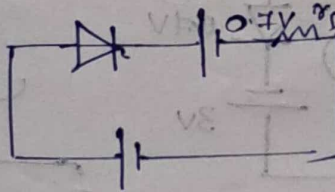
ideal diode



$$I_D = I_S \left(e^{\frac{V_D}{\eta V_T}} - 1 \right)$$

Reverse saturation current.

* practical diode:



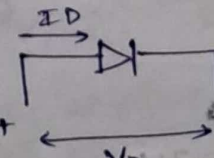
(a short in series with a battery)

* $V_D \rightarrow$ applied forward bias voltage.

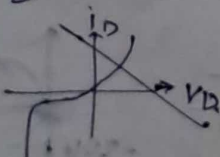
$\eta \rightarrow$ ideality factor $\rightarrow$ function of operating conditions

range from 1 to 2 ($\eta = 1$ assumed unless mentioned otherwise).

$V_T \approx 26 \text{ mV}$ at room temp. $= \frac{kT}{q}$ - Boltzmann const.
 q - charge.
 $\hookrightarrow$ thermal voltage



$V_S - V_D = I_D R_L$

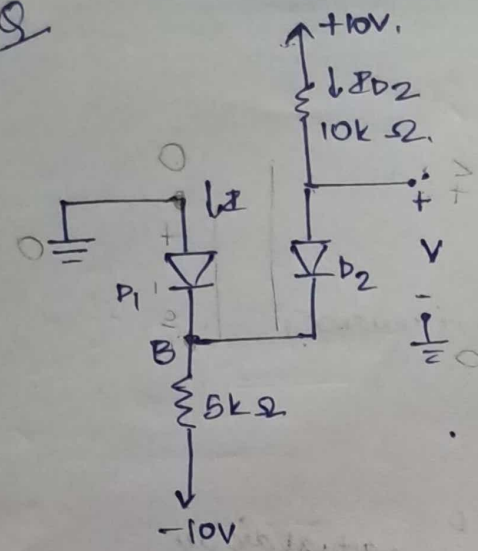


A.S. Seltra & K.C. Smith
- microelectronic circuits

Electronic devices & circuit theory

- R.L. Boylestad & L. Nashelsky

9



Ideal diodes.
assume both diodes are on,
and short them,
and verify.

(if they are not ideal,
replace them with a 0.7V/0.3V battery
and verify)

if its not correct (say current is flowing from
cathode to anode)

check other combination

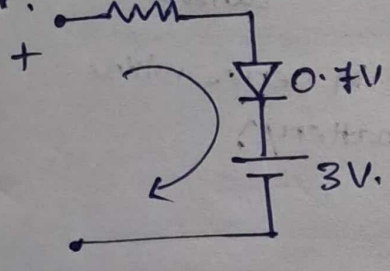
for D_1 is off & D_2 is on, etc.

opened / shorted

check if anode voltage is
higher than cathode voltage.

lec-2
29/9/23

(AC input) - (4 cases) non-ideal diode



$$V_i < 3.7V \Rightarrow \text{off}$$

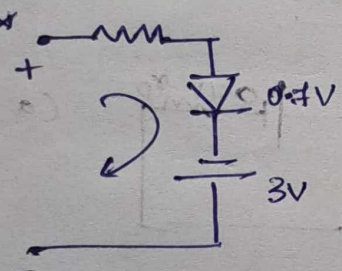
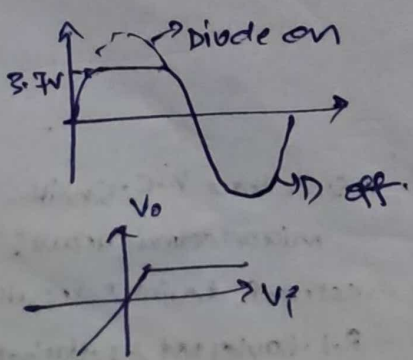
$$\Rightarrow V_0 = V_i$$

$$V_i = IR + 3.7$$

$$V_i > 3.7V \Rightarrow \text{on}$$

$$V_0 = 3.7V$$

(output
clipped at 3.7V)



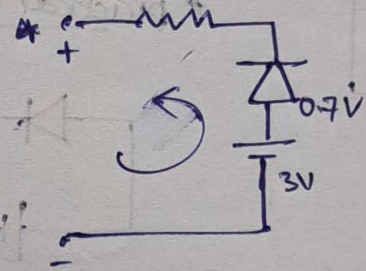
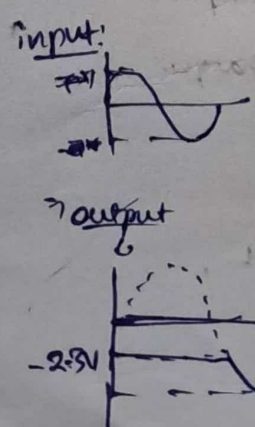
$$V_i' > 0.7 - 3$$

$$= -2.3V$$

$$\Rightarrow \text{Don.}$$

$$V_0 = 3$$

$$V_0 = -2.3V$$



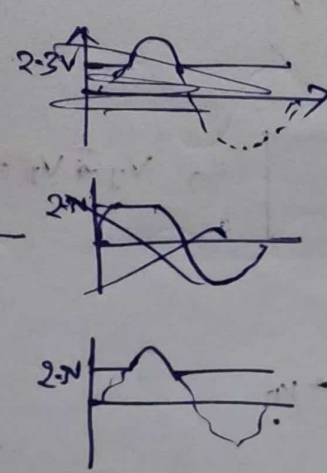
$$-V_i + 3 - 0.7 - IR = 0$$

$$V_i = 2.3 - IR$$

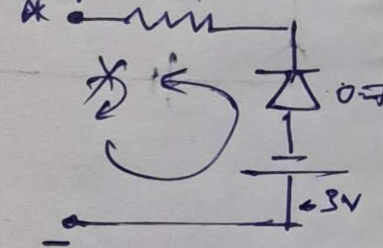
$$V_i > 2.3V$$

$$\Rightarrow \text{D off}$$

$$V_i < 2.3V \Rightarrow \text{Don.}$$



(never change diode
polarity)



$$V_i' > -3 - 0.7 - IR$$

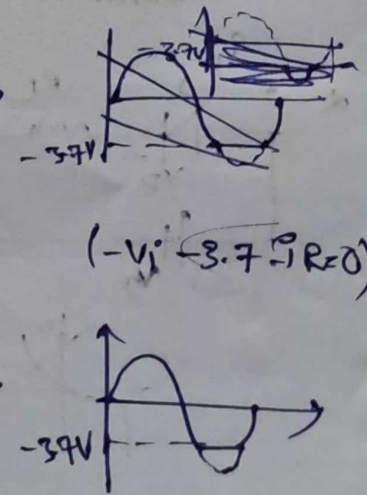
$$\Rightarrow -3.7V - IR$$

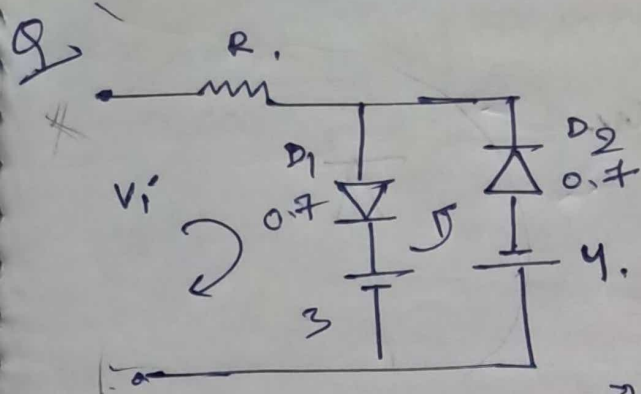
$$\Rightarrow V_i' > -3.7V \text{ D off}$$

$$\Rightarrow \text{D off}$$

$$V_i' < -3.7V \text{ Don.}$$

$$\text{D off}$$

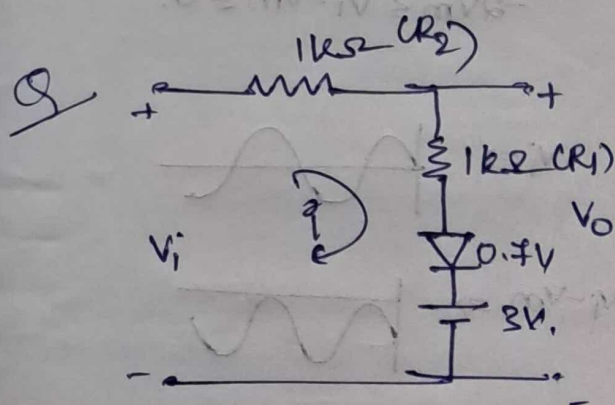
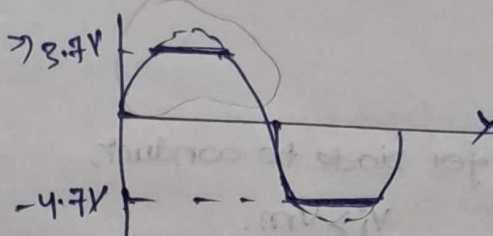




Case 1:
 $\Rightarrow D_1$ conducts & D_2 doesn't
 $\Rightarrow V_i > 3.7V$.
 Den

$\Rightarrow V_i < 3.3V$ off
 $V_i < -4.7V$

$\Rightarrow D_2$ off on



$V_i > 3.7V \Rightarrow D$ on

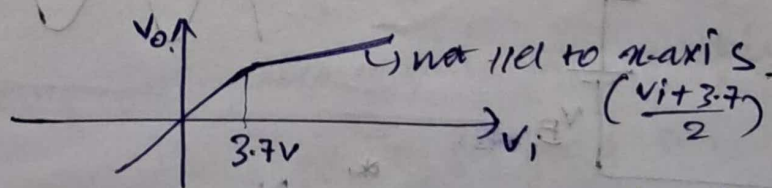
but it won't be clipped exactly at 3.7V

$\Rightarrow$ voltage drop across the resistor R_1 should also be considered.

$$I = \frac{V_i - 3.7}{2 \times 1000}$$

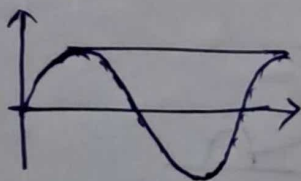
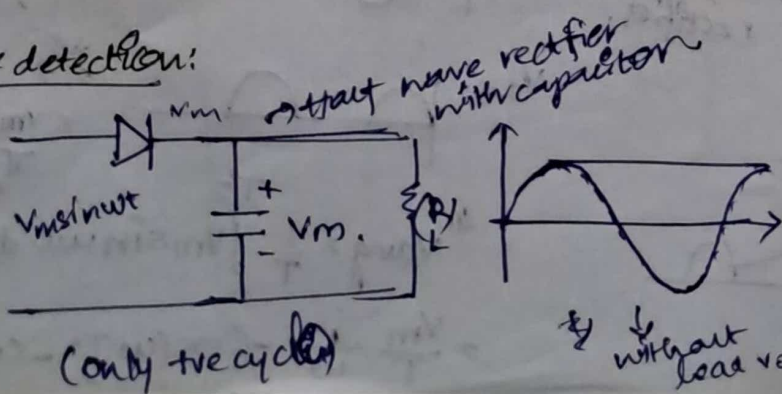
$$V_o = 3.7 + \left(\frac{V_i - 3.7}{2k} \right) \times 1k = \frac{V_i + 3.7}{2} \quad (\text{when } D \text{ is on})$$

D is off $\Rightarrow V_o = V_i$

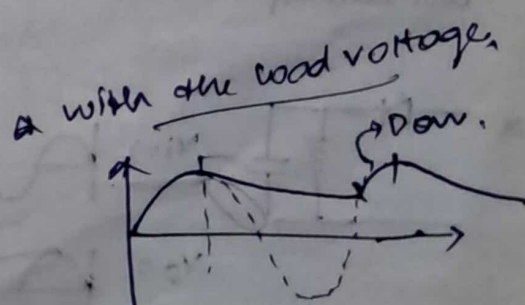


2) $\Rightarrow$ not clipped to a constant voltage, (still a sine wave, amplitude is reduced)
 $\left(\frac{V_{msinwt} + 3.7}{2} = V_o \right)$

A Peak detection:



$\Rightarrow$ without load voltage



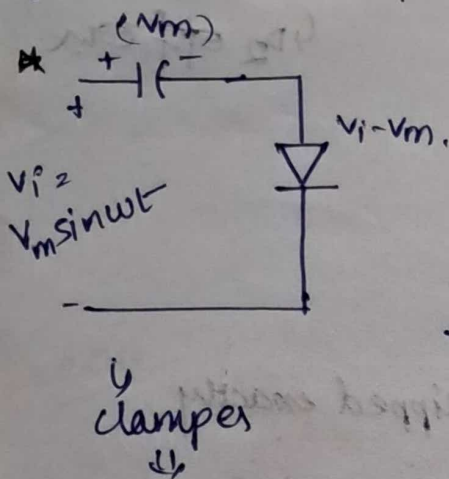
to maintain $V_m \approx V_m e^{-t/RC}$

RC should be very very high.

$$(RC \gg t)$$

06/10/23

CLAMPING



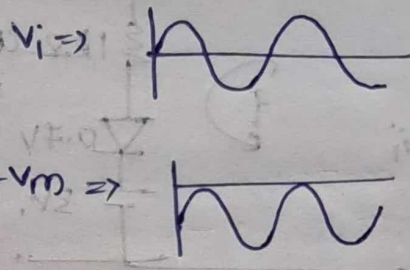
for diode to conduct,
 $V_i > V_m$.
but it doesn't happen.

$$-V_m \leq V_i \leq V_m$$

$$-2V_m \leq V_i - V_m \leq 0$$

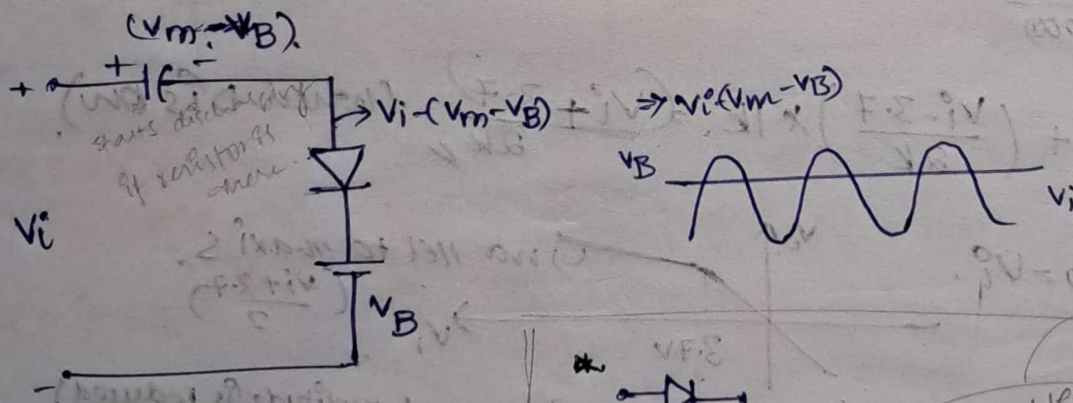
At start, diode conducts.

then instantaneously cap. gets charged and diode opens.

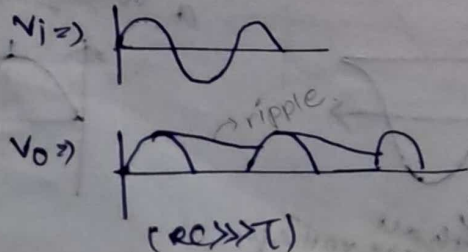
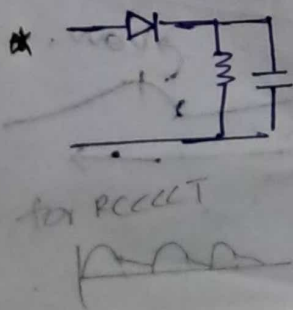
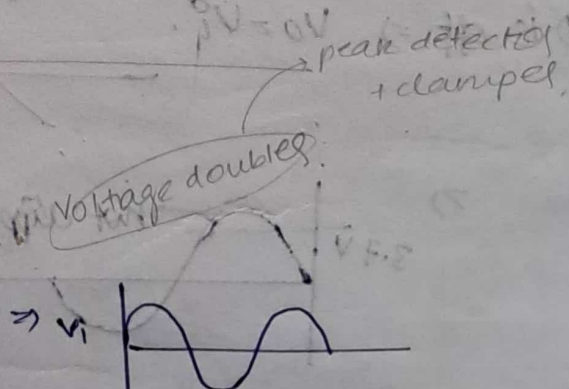
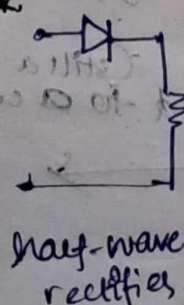


It does not change the shape of the curve but changes the position. (level)

*



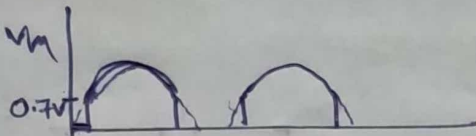
cap. will charge to $V_m - V_B$. diode will turn off.



$$V_{avg} = \frac{1}{T} \int_0^{T/2} V_m \sin \omega t dt$$

$$= \frac{V_m}{T} \cdot \frac{1}{\omega} (-\cos(\omega \cdot \frac{T}{2}) - \cos 0)$$

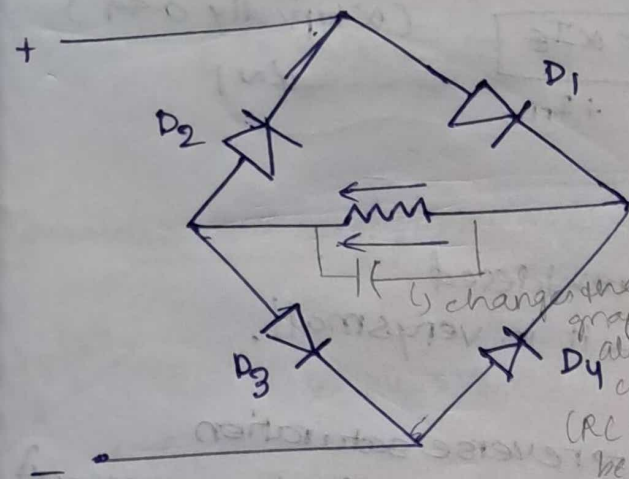
for real diode $\Rightarrow V_m$



$\Rightarrow$ but avg value remains $\frac{V_m}{\pi}$
 (it'll slightly drop but we approx it to $\frac{V_m}{\pi}$)

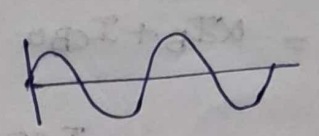
for ideal diode $\Rightarrow \frac{V_m}{\pi}$

Full wave rectifier:

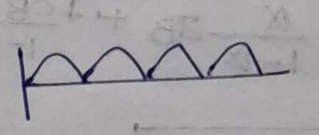


$D2 \& D4 \Rightarrow$ -ve half cycle conducting
 $D1 \& D3 \Rightarrow$ +ve half cycle conducting

$V_i \Rightarrow$



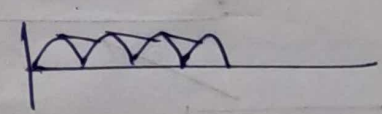
$V_o \Rightarrow$



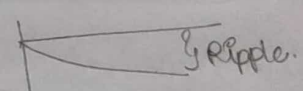
changes the graph to almost const. value
 (RC value should be large)

$V_{avg} = \frac{2V_m}{\pi}$

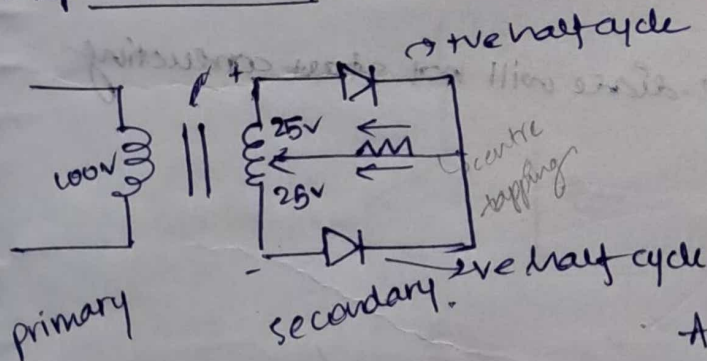
with cap



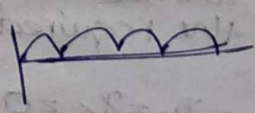
RC filter $\Rightarrow$



Step down transformer:



$V_o \Rightarrow$



like a full wave rectifier

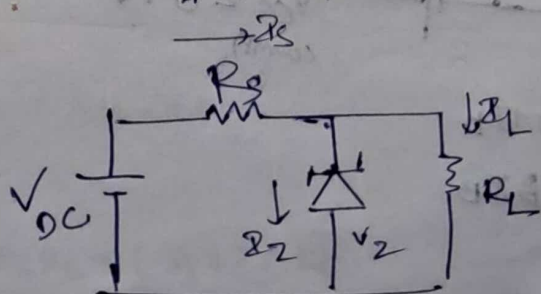
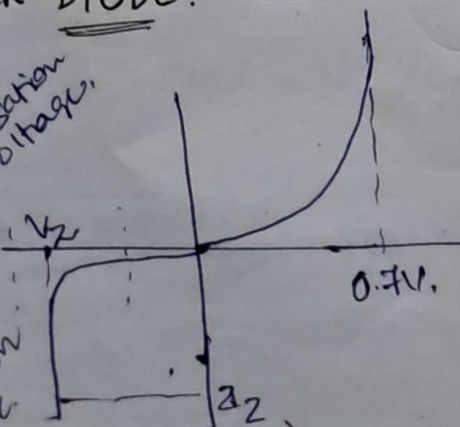
Add a capacitor for good DC.

ZENER DIODE:

(always reverse biased: input = DC)

used for stabilisation of voltage

conducts only at breakdown voltage

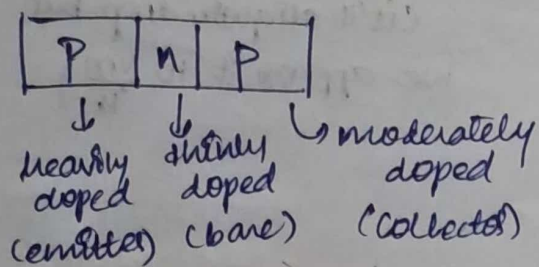


say, breakdown voltage = 5V

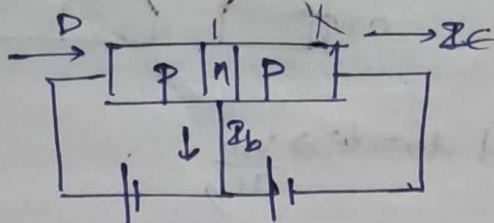
$V_Z = I_Z R_L; R_S = R_Z + R_L$

$\therefore R_S = \frac{V_{DC} - V_Z}{I_Z}$

At. 10/10/23 Bipolar junction transistors



=>



Forward bias

Reverse bias

$$I_C = \alpha(I_B + I_E) + I_{CBO}$$

$$I_E = I_B + I_C$$

$$I_C = \alpha I_E + I_{CBO}$$

(α typically 0.99)
 $\alpha \approx 1$

$$I_C(1 - \alpha) = \alpha I_B + I_{CBO}$$

$$\Rightarrow I_C = \frac{\alpha}{1 - \alpha} I_B + \frac{I_{CBO}}{1 - \alpha}$$

not considered as it is very small.

$$\Rightarrow I_C = \frac{\alpha}{1 - \alpha} I_B$$

$I_{CBO} \Rightarrow$ reverse saturation current (collector current)

$$\star I_C = \beta I_B \Rightarrow I_E = (1 + \beta) I_B$$

The higher the β , the more the current is collected.

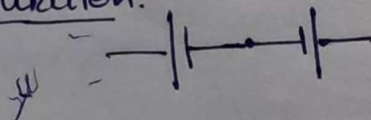
(certain voltage)

If voltage is less than the required, the diode will not start conducting

$$\Rightarrow I_E = I_B = I_C = 0$$

$\Rightarrow$ 'cut-off'.

* Saturation:



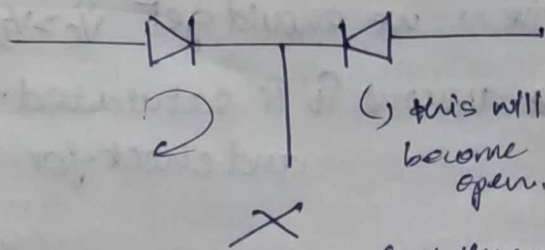
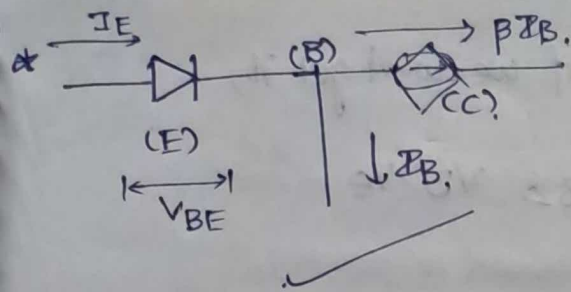
$$V_{CE} \approx 0.1V$$

(comp.)

$$I_E = I_B + I_C$$

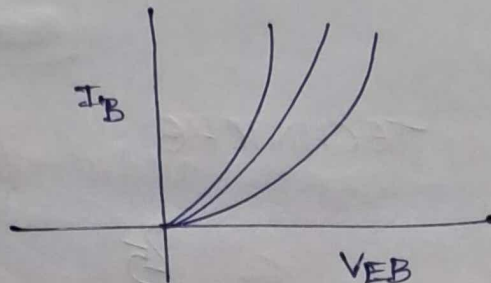
$$I_C \approx \beta I_B$$

$$\beta = \frac{I_C}{I_B}$$



$$I_E = I_{E0} \left(e^{\frac{V_{BE}}{V_T}} - 1 \right)$$

$$I_B = \frac{I_{E0}}{\beta + 1} \left(e^{\frac{V_{BE}}{V_T}} - 1 \right)$$



(this will become open.
All the current will go to base but base is lightly doped so it'll not work)

Common emitter configuration

16/10/23

$$V_C > V_B > V_E$$

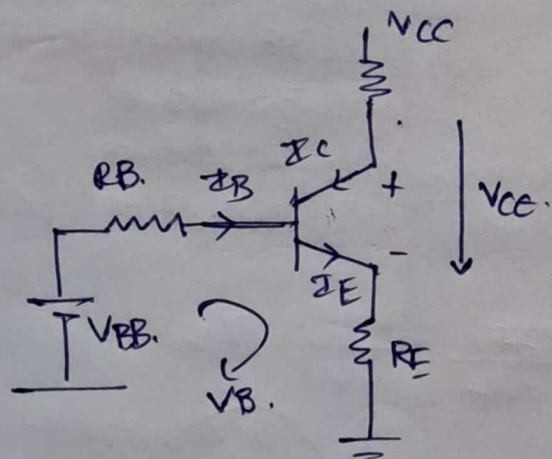
Active:

$$I_C = \beta I_B, I_E = (\beta + 1) I_B$$

Saturation:

$$V_{CE}(\text{sat}) \approx 0.1 \text{ V}$$

$$V_C < V_B, V_B > V_E$$



$$I_C = \beta I_B: V_C = V_{CC} - I_C R_C$$

$$V_E = I_E R_E$$

$$V_{BB} - V_{BE} = I_B R_B + (\beta + 1) I_B R_E$$

$$I_B = \frac{V_{BB} - V_{BE}}{R_B + (\beta + 1) R_E}$$

$$V_{CE} = V_C - V_E; V_B = V_{BB} - I_B R_B$$

→ For saturation,

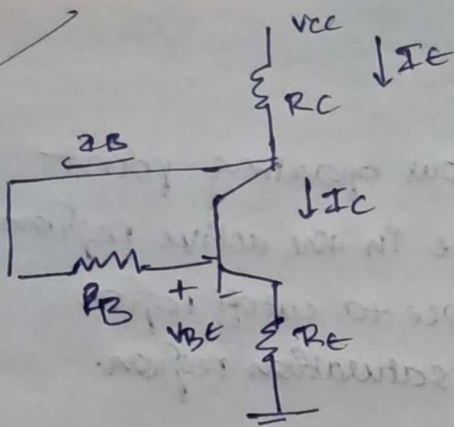
$$V_{BB} - V_{BE} = I_B R_B + (I_B + I_C) R_E$$

$$V_{CC} - V_{CE}(\text{sat}) = I_C R_C + (I_B + I_C) R_E$$

⇒ say it's active ⇒ we should get $V_C > V_B > V_E$ ⇒ if we don't get it,
~~it's~~ assume it is saturated.
and check for $V_C < V_B$ & $V_B > V_E$.

(V_C, V_B, V_E should be close by)

20/10/23

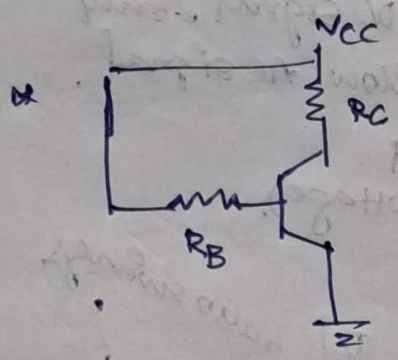
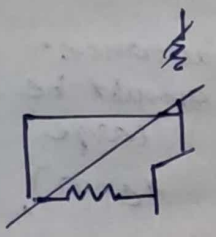


$$V_{CC} - I_E R_C - I_B R_B - V_{BE} - I_E R_E = 0.$$

$$I_C = \frac{\beta(V_{CC} - V_{BE})}{R_B + (R_C + R_E)(\beta + 1)}$$

$$\sim \frac{V_{CC} - V_{BE}}{R_C + R_E} \quad (\beta \text{ very very large}),$$

$$(\beta \sim \beta + 1)$$

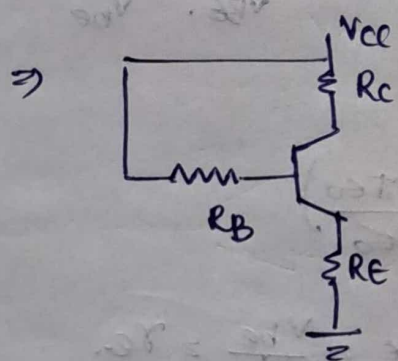


$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$I_C = \frac{\beta(V_{CC} - V_{BE})}{R_B}$$

$\Rightarrow I_C$ varies with β , unlike that of the I_C in the above circuit.

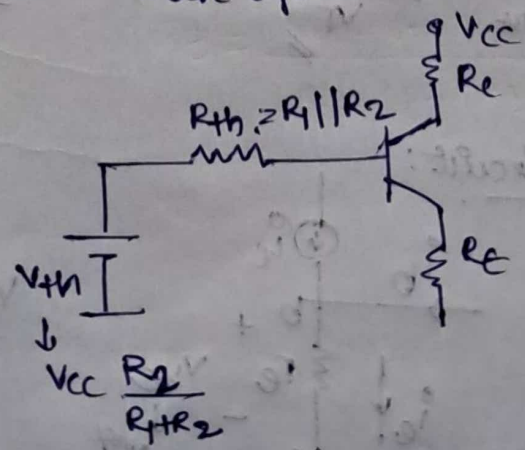
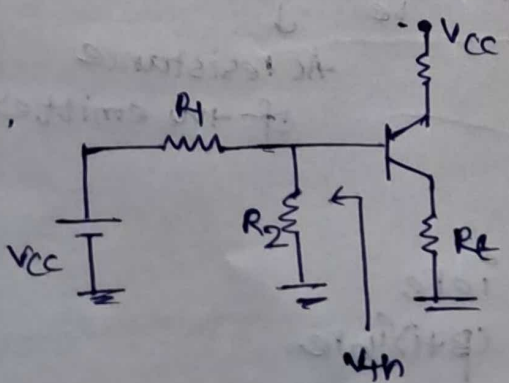
To stabilize this, we can add R_E to the circuit.



$$I_C = \frac{\beta(V_{CC} - V_{BE})}{R_B + (\beta + 1)R_E}$$

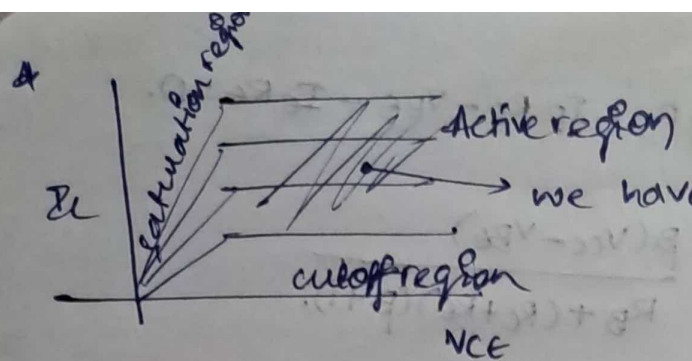
DC voltage input

$\Rightarrow I_C$ is less sensitive to change in β than before.

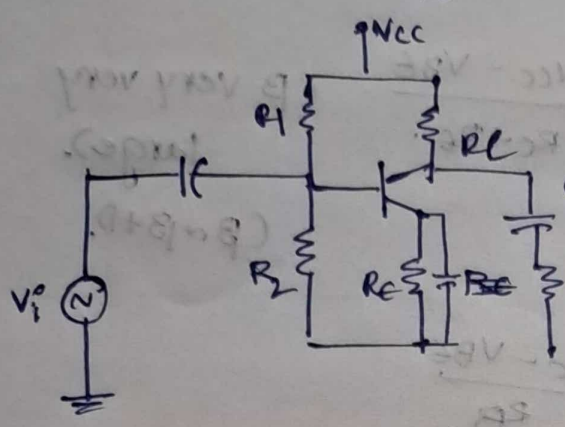


Voltage divider.

$$\Rightarrow V_{th} - V_{BE} = I_B R_{th} + (\beta + 1) I_B R_E \Rightarrow I_B = \frac{V_{th} - V_{BE}}{R_{th} + (\beta + 1) R_E}$$



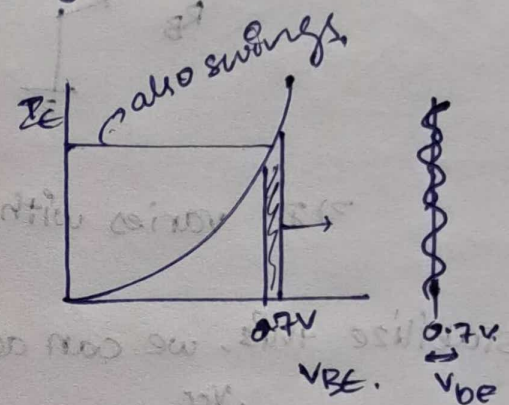
we have to choose our operating point somewhere in the active region - not close to cutoff region or saturation region.



capacitor added to block the DC signal & only allow AC signal. (Amplified input voltage).

(capacitance should be large $X \sim 10$),

$I_E + i_e$ $I_C + i_c$ $I_B + i_b$
 $V_E + v_e$ $V_C + v_c$ $V_B + v_b$



$$I_E \approx I_{E0} \left(e^{\frac{V_{BE}}{V_T}} - 1 \right)$$

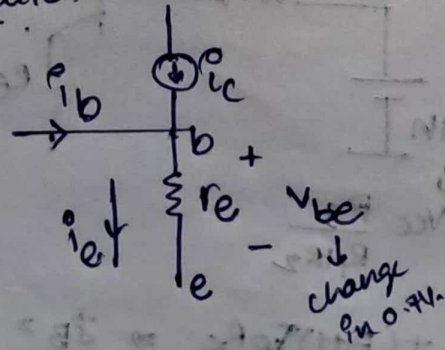
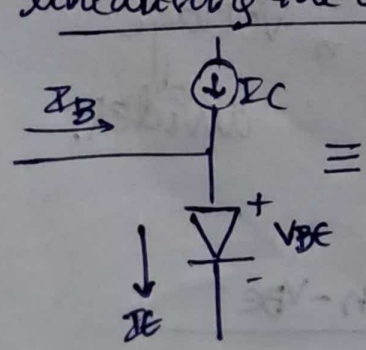
$$\frac{i_e}{v_{be}} \approx \frac{dI_E}{dV_{BE}} = I_{E0} \cdot \frac{1}{V_T} e^{\frac{V_{BE}}{V_T}} = \frac{I_{E0}}{V_T} \left(\frac{I_E + I_{E0}}{I_{E0}} \right)$$

$$\Rightarrow \frac{dI_E}{dV_{BE}} \approx \frac{I_E + I_{E0}}{V_T}$$

$$\frac{dV_{BE}}{dI_E} = \frac{v_{be}}{i_e} = r_e$$

AC resistance of the emitter.

linearizing the circuit:



$$V_{be} = i_e r_e = (\beta + 1) i_b r_e \approx (\beta + 1) r_e i_b$$

*

